

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250278129

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

SASAHARA; NAOTO

INFORMATION PROCESSING APPARATUS CAPABLE OF SUPPRESSING POWER CONSUMPTION IN THE INFORMATION PROCESSING APPARATUS ACCORDING TO USAGE STATE OF THE INFORMATION PROCESSING APPARATUS, CONTROL METHOD FOR INFORMATION PROCESSING APPARATUS, AND STORAGE MEDIUM

Abstract

An information processing apparatus capable of suppressing power consumption according to a usage state is provided. The information processing apparatus includes a power control unit that switches between a first power state and a second power state, an operation section that performs a selection operation that selects enabling or disabling use of the USB device, and a determining unit that determines whether the information processing apparatus is in a connected state or in an unconnected state. The determining unit, in a case where the USB device is connected to the information processing apparatus and use of the USB device is possible and disabling the use of the USB device has been selected by the operation section, determines that the information processing apparatus is in the unconnected state. The power control unit, when being determined that the information processing apparatus is in the unconnected state, switches to the second power state.

Inventors: SASAHARA; NAOTO (Ibaraki, JP)

Applicant: CANON KABUSHIKI KAISHA (Tokyo, JP)

Family ID: 94869698

Appl. No.: 19/057241

Filed: February 19, 2025

Foreign Application Priority Data

JP	2024-031146	Mar. 01, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: G06F1/3246 (20190101); H04N1/00 (20060101)

U.S. Cl.:

CPC G06F1/3246 (20130101); H04N1/00904 (20130101);

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to an information processing apparatus, a control method for the information processing apparatus, and a storage medium.

Description of the Related Art

[0002] An image forming apparatus can be configured in a plurality of standby states. The standby states include, for example, a ready state, a sleep state, and an off state. The ready state is a state in which the image forming apparatus is waiting for input of an instruction for a copy function, a print function, or the like, while the temperature inside the image forming apparatus is rising. The sleep state is a standby state in which the image forming apparatus is not used by a user and power consumption (standby power) is reduced. The off state is a state in which the image forming apparatus is not capable of being used immediately, but the power consumption is the lowest. In addition, the image forming apparatus has a power saving function for transitioning from each standby state to a lower power consumption state. This power saving function enables each component (each part) of the image forming apparatus to operate while the image forming apparatus is configured in a standby state with low power consumption. For example, Japanese Laid-Open Patent Publication (kokai) No. 2015-133636 describes an apparatus that performs a specified display in the case that no operation is performed for a specified automatic stop time, and further turns off a power supply of the apparatus (i.e., automatically shuts down the power supply of the apparatus) in the case that no operation is accepted. In the apparatus disclosed in Japanese Laid-Open Patent Publication (kokai) No. 2015-133636, it is possible to suppress power consumption by automatic shutdown. In addition, in the apparatus disclosed in Japanese Laid-Open Patent Publication (kokai) No. 2015-133636, it is possible to record information regarding the automatic shutdown and change (manage) the time for the automatic shutdown.

[0003] However, in the apparatus disclosed in Japanese Laid-Open Patent Publication (kokai) No. 2015-133636, for example, in the case of determining a connected state (e.g., an available state) of a USB device, there is a risk that the determination may be made simply based on whether the USB device has been connected to a USB port of the apparatus. In this case, even if there is no intention to use the USB device, it is determined that the USB device is in the connected state and supplying power to the USB device will be performed, resulting in a waste of power consumption. As described above, with the apparatus disclosed in Japanese Laid-Open Patent Publication (kokai) No. 2015-133636, depending on the usage state of the apparatus, the effect of reducing power consumption may not be obtained.

SUMMARY OF THE INVENTION

[0004] The present invention provides an information processing apparatus capable of suppressing power consumption in the information processing apparatus according to a usage state of the information processing apparatus. According to embodiments of the invention there is provided a control method for the information processing apparatus, a storage medium, and a program.

[0005] Accordingly, the present invention provides an information processing apparatus which is configurable in a first power state in which the information processing apparatus is supplied with power, and a second power state in which power consumption is lower than that of the first power state, wherein the information processing apparatus, to which a USB device is attachably and detachably connected, comprises a power control unit configured to switch between the first power state and the second power state, an operation section configured to perform a selection operation that selects enabling the use of the USB device by the information processing apparatus or disabling the use of the USB device by the information processing apparatus, and a determining unit configured to determine whether the information processing apparatus is in a connected state in which the USB device is connected to the information processing apparatus, or in an unconnected state in which the USB device is not connected to the information processing apparatus. The determining unit is configured, in a case where the USB device is connected to the information processing apparatus and the use of the USB device by the information processing apparatus is possible and disabling the use of the USB device has been selected by the operation section, to determine that the information processing apparatus is in the unconnected state. The power control unit is configured, in a case where the determining unit determines that the information processing apparatus is in the unconnected state, to switch to the second power state.

[0006] According to the present invention, it is possible to suppress the power consumption in the information processing apparatus according to the usage state of the information processing apparatus.

[0007] Further features of the present invention will become apparent from the following description of example embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a block diagram that shows an example of a hardware configuration of an image forming system that includes an image forming apparatus in the case of having been applied as an information processing apparatus.

[0009] FIG. 2 is a block diagram that shows an example of a hardware configuration of a controller included in the image forming apparatus.

[0010] FIG. 3 is a block diagram that shows an example of a configuration for power control of the image forming apparatus.

[0011] FIG. 4 is a block diagram that shows an example of a hardware configuration of a USB device controller, the USB device controller being included in the image forming apparatus and/or its peripherals.

[0012] FIG. 5 is a diagram that shows an example of a setting screen displayed on an operation unit, the operation unit being included in the image forming apparatus.

[0013] FIG. 6 is a flowchart that shows a startup processing of the USB device controller when the image forming apparatus is started.

[0014] FIG. 7 is a flowchart that shows a processing for switching between enabling and disabling a USB device function on the setting screen shown in FIG. 5.

[0015] FIG. 8 is a flowchart that shows an offline timer processing for automatically shutting down the image forming apparatus.

[0016] FIG. 9 is a flowchart that shows a detailed process performed in a step S804 of the flowchart shown in FIG. 8.

DESCRIPTION OF THE EMBODIMENTS

[0017] The present invention will now be described in detail below with reference to the accompanying drawings showing embodiments thereof.

[0018] Hereinafter, an example embodiment of the present invention will be described in detail with reference to the drawings. However, the configuration described in the following embodiment is merely an example, and the scope of the present invention is not limited by the configuration described in the following embodiment. For example, each unit (each component) constituting the present invention is able to be replaced with a unit (a component) with any configuration that is capable of performing the same function. In addition, any component(s) may be added.

[0019] FIG. 1 is a block diagram that shows an example of a hardware configuration of an image forming system that includes an image forming apparatus in the case of having been applied as an information processing apparatus. As shown in FIG. 1, an image forming system **1000** includes an image forming apparatus **101**, a computer **109**, and a finisher device **150**. In the present embodiment, the image forming apparatus **101** is a multi-function peripheral (an MFP) that has a print function, a scanner function, a data communication function, etc., but is not limited to be an MFP. In addition, the image forming apparatus **101** is configured to be able to be in a first power state in which the image forming apparatus **101** is supplied with power and is operated, and a second power state in which power consumption is lower than that of the first power state. The power states of the image forming apparatus **101** include, for example, a normal state, a ready state, a sleep state, an off state, and a shutdown state. “The normal state” refers to a state in which it is possible to quickly execute desired functions such as the print function, the scanner function, the data communication function, etc. “The ready state” refers to a state in which the image forming apparatus **101** is waiting for input of an instruction for a copy function, the print function, or the like, while the temperature inside the image forming apparatus **101** is rising. “The sleep state” refers to a standby state in which the image forming apparatus **101** is not used by a user and the power consumption (standby power) is reduced. “The off state” refers to a state in which the image forming apparatus **101** is not capable of being used immediately, but the power consumption is the lowest compared to the ready state and the sleep state (the least amount of power is used compared to the ready state and the sleep state). “The shutdown state” refers to a state in which a power supply **301** (see FIG. 3) of the image forming apparatus **101** has been turned off, that is, power supplying from the power supply **301** has been cut off. In the present embodiment, for example, the normal state, the ready state, the sleep state, and the off state are treated as the first power state, and the shutdown state is treated as the second power state, but the present invention is not limited to this kind of treatment.

[0020] The image forming apparatus **101** includes a scanner device **102**, a controller (a control device) **103**, a printer device **104**, an operation unit (a display unit) **105**, a hard disk (a hard disk drive (an HDD)) **106**, and a facsimile device **107**. The scanner device **102** optically reads an image from a document and converts it into a digital image. In the present embodiment, the scanner device **102** includes a document feeding unit (a DF unit) **121** and a scanner unit **122**. The document feeding unit **121** is able to automatically and sequentially replace bundles of documents. The scanner unit **122** optically scans the document and converts it into image data (the digital image) of the document. This image data is transmitted to the controller **103**. The printer device **104** prints the image data on sheets (printing sheets). In the present embodiment, the printer device **104** includes a marking unit **141**, a sheet feeding unit **142**, and a sheet discharging unit **143**. The sheet feeding unit **142** sequentially supplies sheets one by one from a bundle of sheets to the marking unit **141**. The marking unit **141** prints the image data on the sheets that have been supplied from the sheet feeding unit **142**. The sheet discharging unit **143** discharges the printed matter (the printed sheets) on which the image data has been printed.

[0021] The operation unit **105** includes a touch panel, hard keys, and the like for accepting settings for the image forming apparatus **101** from the user and displaying the processing status of the image forming apparatus **101**. In addition, as a function for accepting the settings for the image forming apparatus **101** from the user, for example, the operation unit **105** is capable of being provided with setting switches for enabling or disabling a given setting (e.g., a setting switch may

be provided to enable or disable each of the suspend, hibernation, and hybrid sleep settings). In addition, the operation unit **105** is able to refer to, from an application, flag(s) that have been set. The hard disk **106** is a nonvolatile storage device such as a solid state drive (an SSD) or an embedded multi-media card (an eMMC) that stores, for example, the image data, control programs, etc. The control programs include, for example, programs or the like for causing the controller **103** (a computer) to execute the respective units of the image forming apparatus **101** and the respective means of the image forming apparatus **101** (a control method for the information processing apparatus). The facsimile device **107** is able to transmit various types of information to a telephone line **160** or the like, and is able to receive various types of information from the telephone line **160** or the like. The controller **103** is a computer that is communicably connected to the scanner device **102**, the printer device **104**, the operation unit **105**, the hard disk **106**, and the facsimile device **107**, and controls the operations of these units (the scanner device **102**, the printer device **104**, the operation unit **105**, the hard disk **106**, and the facsimile device **107**). The controller **103** is able to execute various kinds of jobs such as a print job by controlling the operations of these units.

[0022] Examples of jobs (functions) that can be executed by the image forming apparatus **101** include the copy function, an image transmission function, an image storage function, and an image printing function. “The copy function” is a function for recording image data that has been read by the scanner device **102** onto the hard disk **106** and for performing printing of the image data by the printer device **104**. “The image transmission function” is a function for transmitting the image data that has been read by the scanner device **102** to the computer **109** via a local area network (a LAN) **108**. “The image storage function” is a function for recording the image data that has been read by the scanner device **102** onto the hard disk **106**, and for performing image data transmission or image data printing as necessary. “The image printing function” is a function for analyzing language data (for example, a page description language) that has been transmitted from the computer **109** and printing the language data by the printer device **104**.

[0023] The image forming apparatus **101** is communicably connected to the computer **109** via the LAN **108**. As a result, the image forming apparatus **101** is able to receive various kinds of jobs from the computer **109**, and is able to transmit and receive image data to and from the computer **109**. It should be noted that although the number of computers connected to the image forming apparatus **101** is one in the configuration shown in FIG. 1, the number of computers connected to the image forming apparatus **101** is not limited to one and may be, for example, two or more. In addition, the image forming apparatus **101** and the computer **109** are also capable of being communicably connected to each other via a universal serial bus (a USB) **110**. In this case, the image forming apparatus **101** functions as a USB device. The finisher device **150** performs various kinds of processes, such as sorting, stapling, punching holes, and cutting, with respect to the printed matter (the printed sheets) outputted from the sheet discharging unit **143** of the printer device **104** of the image forming apparatus **101**.

[0024] FIG. 2 is a block diagram that shows an example of a hardware configuration of the controller included in the image forming apparatus. As shown in FIG. 2, the controller **103** includes a main board **200** and a sub-board **220**. The main board **200** is a general-purpose CPU system, and includes a boot ROM **202**, a bus controller **204**, a nonvolatile memory **205**, a disk controller **206**, and a flash disk **207**. In addition, the main board **200** includes a USB host controller **208**, a USB device controller **210**, a network controller **211**, a real time clock (an RTC) **212**, a watchdog timer (a WDT) **230**, a central processing unit (a CPU) **340**, and a memory **341**. These modules on the main board **200** are communicably connected to each other.

[0025] The boot ROM **202** stores a boot program. The bus controller **204** has a bridge function with an external bus. The nonvolatile memory **205** is a memory in which data is not lost even in the case that the power supply has been cut off. The disk controller **206** controls a storage device. The flash disk **207** is a relatively small-capacity storage device made up of a semiconductor device, and is made up of, for example, an SSD or an eMMC. The USB host controller **208** functions as a USB

host. The USB device controller **210** is a controller that enables the image forming apparatus **101** to function as a USB device. The network controller **211** transmits and receives data to and from the computer **109** via the LAN **108**. It should be noted that although the number of network controllers **211** arranged in the configuration shown in FIG. 2 is one, the number of network controllers **211** arranged is not limited to one, and may be, for example, two or more. In addition, the network controller **211** may be capable of controlling a wireless LAN. In addition, the network controller **211** may be connected to the USB host controller **208**. The RTC **212** sets a current time and a return time. The watchdog timer **230** resets the controller **103**. The CPU **340** controls the modules such as the boot ROM **202**, etc., that is, controls the entire main board **200**. It should be noted that the CPU **340** includes a large number of hardware components, such as a chip set, a bus bridge, and a clock generator. The memory **341** is used as a working memory for the CPU **340**. A USB memory **209**, the operation unit **105**, the hard disk **106**, etc. are communicably connected to the main board **200**.

[0026] The sub-board **220** is configured with a relatively small general-purpose CPU system and image processing hardware. The sub-board **220** includes a CPU **221**, a memory **223**, a bus controller **224**, a nonvolatile memory **225**, device controllers **226**, and an image processor **227**, which are connected to each other so as to be able to communicate with each other. The CPU **221** controls the modules such as the memory **223**, etc., that is, controls the entire sub-board **220**, and the CPU **221** also controls the facsimile device **107**. It should be noted that similar to the CPU **340**, the CPU **221** includes a large number of hardware components, such as a chip set, a bus bridge, and a clock generator. The memory **223** is used as a working memory for the CPU **221**. The bus controller **224** has a bridge function with an external bus. The nonvolatile memory **225** is a memory in which data is not lost even in the case that the power supply has been cut off. In the present embodiment, the two device controllers **226** are provided. The scanner device **102** is communicably connected to one of the two device controllers **226**. As a result, it is possible to perform transmission of image data from the scanner device **102** to the sub-board **220**. In addition, the printer device **104** is communicably connected to the other of the two device controllers **226**. As a result, it is possible to perform transmission of image data from the sub-board **220** to the printer device **104**. The image processor **227** performs digital image processing in real time.

[0027] FIG. 3 is a block diagram that shows an example of a configuration for power control of the image forming apparatus. As shown in FIG. 3, as a power control configuration, the image forming apparatus **101** includes the power supply **301**, a power supply control unit **303**, a first power switch **310**, a second power switch **311**, a third power switch **312**, a fourth power switch **313**, and a power switch **110**. The power supply **301**, the first power switch **310**, the second power switch **311**, the third power switch **312**, the fourth power switch **313**, and the power switch **110** are connected to the power supply control unit **303**. In addition, as described above, the image forming apparatus **101** includes the scanner device **102**, the controller **103**, the printer device **104**, and the operation unit **105**. The scanner device **102** is connected to the power supply **301** and the power supply control unit **303** via the third power switch **312**. The controller **103** is connected to the power supply **301** and the power supply control unit **303** via the first power switch **310**. The printer device **104** is connected to the power supply **301** and the power supply control unit **303** via the fourth power switch **313**. The operation unit **105** is connected to the power supply **301** and the power supply control unit **303** via the second power switch **311**. The CPU **340** is built in the controller **103**. A CPU **307** is built in the printer device **104**. A CPU **305** is built in the operation unit **105**.

[0028] The power supply control unit **303** is constantly supplied with power from the power supply **301** via a power supply line. It should be noted that when the power supply is turned off, only the power supply control unit **303** is energized and the power control is performed. This makes it possible to reduce the power consumption to a minimum. In the image forming apparatus **101**, when the power switch **110** which is a main power switch has been operated, that is, has been pressed down, the power supply control unit **303** detects the operation of the power switch **110**. Then, the power supply control unit **303** controls the first power switch **310** to supply power to the

CPU **340** of the controller **103**, that is, controls the first power switch **310** to supply power from the power supply **301** to the CPU **340** of the controller **103**. Similarly, the power supply control unit **303** controls the second power switch **311** to supply power to the CPU **305** of the operation unit **105**, controls the third power switch **312** to supply power to the scanner device **102**, and controls the fourth power switch **313** to supply power to the CPU **307** of the printer device **104**.

[0029] It should be noted that the CPU **340** of the controller **103** is able to control the second power switch **311** by a notification to the power supply control unit **303** to supply power individually to the CPU **305** of the operation unit **105**. Similarly, the CPU **340** is able to, by a notification to the power supply control unit **303**, control the third power switch **312** to supply power individually to the scanner device **102**, or control the fourth power switch **313** to supply power individually to the printer device **104**. In addition, the power supply control unit **303** or the CPU **340** is also able to perform power supplying control with respect to the marking unit **141**, the sheet feeding unit **142**, and the sheet discharging unit **143** of the printer device **104**. In addition, power supplying for each block is also able to be realized by, for example, configuring the first power switch **310** with two systems, and in the sleep state, turning off only one relay switch connected to the block that turns off the power supply **301**, and leaving the other relay switch on. In addition, in the shutdown state, the relay switches of both systems are turned off. In this case, the power supply control signal is not a binary signal but a multi-value control signal according to the energized state. In each power state, such as the sleep state or the shutdown state, power supplying is performed through such control. In addition, in this control, the power supply control unit **303** may perform multi-value control of the first power switch **310** to supply power to each block connected to the controller **103**. In addition, in this control, the CPU **340** of the controller **103** may perform multi-value control of the first power switch **310** by a notification to the power supply control unit **303** to supply power to each block connected to the controller **103**.

[0030] FIG. **4** is a block diagram that shows an example of a hardware configuration of the USB device controller included in the image forming apparatus and its peripherals. As shown in FIG. **4**, the USB device controller **210** includes a LINK section **401**, a PHY section **402**, and a setting register **403**. The LINK section **401** is a part within the USB device controller **210** to be controlled by the CPU **340**. The PHY section **402** is a part connected to an external USB host (a USB device) **407** via a USB device interface (a USB device I/F) **406**. It should be noted that the USB host **407** is attachably and detachably connected to the USB device I/F **406**. In addition, in the present embodiment, the USB host **407** is the computer **109** or the USB memory **209** that is capable of being connected via the USB **110**, but is not limited to be the computer **109** or the USB memory **209**, and may be, for example, a USB device configured to be connectable via USB such as a USB cable. The setting register **403** stores setting information for functioning as a USB device. For example, communication with the USB host **407**, settings for the USB device controller **210**, etc., are performed by a USB device driver controlled by the CPU **340** executing commands such as writing into the LINK section **401**, reading into the LINK section **401**, etc. A control register **404** outside the USB device controller **210** is a control register that exists so that the CPU **340** is able to reset the USB device controller **210**. When the USB device controller **210** is connected to the USB host **407** via the USB device I/F **406**, the USB device controller **210** detects the power supplying from a VBUS line. Then, the USB device controller **210** notifies the CPU **340** via an interrupt line **405** of an interrupt indicating that the VBUS line is in an ON state. In addition, at this time, data indicating that the VBUS line is in the ON state is stored in the setting register **403**. In addition, depending on the type of the USB host **407**, power supplying is performed from the USB device controller **210** to the USB host **407**.

[0031] FIG. **5** is a diagram that shows an example of a setting screen displayed on the operation unit included in the image forming apparatus. A setting screen (an operation screen) **500** shown in FIG. **5** is displayed on the operation unit **105** of the image forming apparatus **101**. The setting screen **500** includes a setting (an operation section) **501**. By using the setting **501**, a selection

operation for switching between enabling and disabling a USB device function in the image forming apparatus **101** is performed, that is, a selection operation for selecting an “ON” state that enables the use of the USB host **407** in the image forming apparatus **101**, or an “OFF” state that disables the use of the USB host **407** in the image forming apparatus **101** is performed (an operation step). The CPU **340** is able to store the setting state on the setting screen **500** in a storage device (a storage unit) such as the HDD **106**, the nonvolatile memory **205**, or the flash disk **207**. [0032] FIG. **6** is a flowchart that shows a startup processing of the USB device controller when the image forming apparatus is started. As shown in FIG. **6**, in a step **S601**, the CPU **340** reads out the setting state on the setting screen **500** that has been stored in the storage device, and determines whether or not the setting **501** of the setting screen **500** is enabled. As a result of the determination in the step **S601**, in the case of being determined that the setting **501** is enabled (YES in the step **S601**), the startup processing proceeds to a step **S602**. On the other hand, as the result of the determination in the step **S601**, in the case of being determined that the setting **501** is not enabled, that is, that the setting **501** is disabled (NO in the step **S601**), the startup processing ends.

[0033] In the step **S602**, the CPU **340** executes an initialization processing of the USB device controller **210**. The initialization processing is a processing that performs writing descriptor information into the setting register **403**, and the like. By executing this initialization processing, a preparation for performing communication between the USB device controller **210** and the USB host **407** is completed. After the step **S602** is executed, the startup processing proceeds to a step **S603**.

[0034] In the step **S603**, the CPU **340** stores, in the memory **341**, a flag indicating that the preparation for communication between the USB device controller **210** and the USB host **407** is completed, that is, a flag indicating information that the use of the USB host **407** in the image forming apparatus **101** is possible (the USB host **407** is capable of being used in the image forming apparatus **101**). It should be noted that this flag is generated, for example, by the CPU **340** (an information generating unit).

[0035] FIG. **7** is a flowchart that shows a processing for switching between enabling and disabling the USB device function on the setting screen shown in FIG. **5**. As shown in FIG. **7**, in a step **S701**, the CPU **340** accepts an operation change with respect to the setting **501** of the setting screen **500**. This “operation change” refers to changing (switching) from enabled to disabled in the setting **501**, or conversely, changing (switching) from disabled to enabled in the setting **501**.

[0036] In a step **S702**, the CPU **340** stores, in the storage device, the operation change with respect to the setting **501** that has been accepted in the step **S701** (the changed setting value).

[0037] In a step **S703**, the CPU **340** displays a message prompting a restart. This message is displayed, for example, in a region **502** provided below the setting **501** on the setting screen **500**. In addition, the user of the image forming apparatus **101** is able to confirm the message displayed in the region **502** and operate a restart button (not shown) that is displayed in the region **502** together with the message.

[0038] In a step **S704**, the CPU **340** determines whether or not the restart button has been operated and an instruction to restart (a restart instruction) has been accepted. As a result of the determination in the step **S704**, in the case of being determined that a restart instruction has been accepted (YES in the step **S704**), the processing for switching between enabling and disabling the USB device function on the setting screen proceeds to a step **S705**. On the other hand, as the result of the determination in the step **S704**, in the case of being determined that a restart instruction has not been accepted (NO in the step **S704**), the processing for switching between enabling and disabling the USB device function on the setting screen ends.

[0039] In the step **S705**, the CPU **340** causes the image forming apparatus **101** to restart. By this restart, the image forming apparatus **101** enters a state in which the content of the operation change with respect to the setting **501** of the setting screen **500** has been reflected. For example, in the case that the operation change in the step **S701** is changing from enabled to disabled, the image forming

apparatus **101** enters a state in which the USB device function is disabled. In addition, in the case that the operation change in the step **S701** is changing from disabled to enabled, the image forming apparatus **101** enters a state in which the USB device function is enabled.

[0040] FIG. **8** is a flowchart that shows an offline timer processing for automatically shutting down the image forming apparatus. As shown in FIG. **8**, in a step **S801**, the CPU **340** starts the operation of an offline timer (not shown). It should be noted that the offline timer is built into, for example, the main board **200**.

[0041] In a step **S802**, the CPU **340** waits for one second. It should be noted that the waiting time is not limited to one second. After the step **S802** is executed, the offline timer processing proceeds to a step **S803**.

[0042] In the step **S803**, the CPU **340** determines whether or not the offline timer has expired, that is, whether or not the time is up. After the offline timer has expired (YES in the step **S803**), the offline timer processing proceeds to a step **S804**.

[0043] In the step **S804**, the CPU **340** controls the USB device controller **210** and the network controller **211** to determine whether the image forming apparatus **101** is in a USB-connected state or a USB-unconnected state (a determining step). “The USB-connected state” refers to a connected state in which the USB host **407** is connected to the USB device I/F **406**. “The USB-unconnected state” refers to an unconnected state in which the USB host **407** is not connected to the USB device I/F **406**. In this manner, in the present embodiment, the CPU **340** also functions as a determining unit that determines whether the image forming apparatus **101** is in the USB-connected state or the USB-unconnected state. It should be noted that in the image forming apparatus **101**, a part functioning as the determining unit may be provided separately from the CPU **340**. In addition, when it is determined in the step **S804** whether the image forming apparatus **101** is in the USB-connected state or the USB-unconnected state, prior to the determination, waiting for one second (waiting for a predetermined period of time) (the step **S802**). Then, after one second has elapsed, the determination in the step **S804** is made. With such a configuration, for example, the determination in the step **S804** is capable of being made stably with a margin of one second. Then, as a result of the determination in the step **S804**, in the case of being determined that the image forming apparatus **101** is in the USB-connected state (NO in the step **S804**), the offline timer processing proceeds to a step **S806**. On the other hand, as the result of the determination in the step **S804**, in the case of being determined that the image forming apparatus **101** is in the USB-unconnected state (YES in the step **S804**), the offline timer processing proceeds to a step **S805**.

[0044] In the step **S805**, the CPU **340** executes a shutdown processing that switches the power state of the image forming apparatus **101** to the shutdown state (the second power state). It should be noted that it is preferable that the shutdown processing in the step **S805** is executed after, for example, ten seconds have elapsed (after a predetermined period of time has elapsed) since it is determined in the step **S804** that the image forming apparatus **101** is the USB-unconnected state. In this case, for example, a countdown until the shutdown processing is performed is displayed on the operation unit **105**. As a result, it is possible to prevent the user from feeling that the image forming apparatus **101** has suddenly entered the shutdown state.

[0045] In the step **S806**, the CPU **340** resets the offline timer. In addition, the CPU **340** switches the power state of the image forming apparatus **101** to the first power state. In this manner, in the present embodiment, the CPU **340** also functions as a power control unit that performs switching control (a power control step) for switching the power state of the image forming apparatus **101** between the first power state and the second power state. It should be noted that in the image forming apparatus **101**, a part functioning as the power control unit may be provided separately from the CPU **340**.

[0046] FIG. **9** is a flowchart that shows a detailed process performed in the step **S804** of the flowchart shown in FIG. **8**. The flowchart shown in FIG. **9** is a flowchart that shows the details of a connection state determination processing for determining whether the image forming apparatus

101 is in the USB-connected state or the USB-unconnected state. As shown in FIG. 9, in a step **S901**, the CPU **340** determines whether or not the preparation for communication between the USB device controller **210** and the USB host **407** is completed, based on whether or not the flag has been stored in the memory **341** in the step **S603**. As a result of the determination in the step **S901**, in the case of being determined that the preparation for communication between the USB device controller **210** and the USB host **407** is completed (YES in the step **S901**), the connection state determination processing proceeds to a step **S902**. On the other hand, as the result of the determination in the step **S901**, in the case of being determined that the preparation for communication between the USB device controller **210** and the USB host **407** is not completed (NO in the step **S901**), the connection state determination processing proceeds to a step **S905**. As described above, the setting state on the setting screen **500** (the setting state of the setting **501**) has been stored in the storage device. In this case, the determination in the step **S901** may be made based on the setting state on the setting screen **500** that has been stored in the storage device.

[0047] In the step **S902**, the CPU **340** obtains information about the state of the VBUS line that has been stored in the setting register **403**, that is, obtains information regarding the presence or absence of power supplying from the VBUS line to the USB device controller **210**. After the step **S902** is executed, the connection state determination processing proceeds to a step **S903**.

[0048] In the step **S903**, the CPU **340** determines whether or not the VBUS line is in the ON state based on the information about the state of the VBUS line that has been obtained in the step **S902**. As a result of the determination in the step **S903**, in the case of being determined that the VBUS line is in the ON state (YES in the step **S903**), the connection state determination processing proceeds to a step **S904**. On the other hand, as the result of the determination in the step **S903**, in the case of being determined that the VBUS line is not in the ON state, that is, the VBUS line is in an OFF state (NO in the step **S903**), the connection state determination processing proceeds to the step **S905**.

[0049] In the step **S904**, the CPU **340** determines that the image forming apparatus **101** is in the USB-connected state.

[0050] In the step **S905**, the CPU **340** determines that the image forming apparatus **101** is in the USB-unconnected state.

[0051] As described above, in the image forming apparatus **101**, based on whether or not the USB host **407** is connected to the USB device controller **210** (the image forming apparatus **101**) and the setting state of the setting **501**, it is determined whether the image forming apparatus **101** is in the USB-connected state or the USB-unconnected state. Specifically, in the case that the USB host **407** is connected and the use of the USB host **407** in the image forming apparatus **101** becomes possible and enabled has been selected in the setting **501** (YES in the step **S901** and YES in the step **S903**), it is determined that the image forming apparatus **101** is in the USB-connected state. On the other hand, even in the case that the USB host **407** is connected and the use of the USB host **407** in the image forming apparatus **101** becomes possible and when disabled has been selected in the setting **501** (NO in the step **S901**), it is determined that the image forming apparatus **101** is in the USB-unconnected state. In this manner, in the case that disabled has been selected in the setting **501**, the determination of the presence or absence of power supplying to the USB device controller **210** (the USB host **407**) (the step **S903**) is omitted (skipped), and it is determined that the image forming apparatus **101** is in the USB-unconnected state. In addition, also in the case that enabled has been selected in the setting **501** but the USB host **407** is not currently connected to the image forming apparatus **101** (YES in the step **S901** and NO in the step **S903**), it is determined that the image forming apparatus **101** is in the USB-unconnected state.

[0052] In the image forming apparatus **101**, for example, in the case that there is no intention to use the USB host **407** even though the USB host **407** is connected, it is determined that the image forming apparatus **101** is in the USB-unconnected state. As a result, it is possible to supply power to the USB host **407**, and accordingly, it is possible to suppress the waste of the power

consumption. Therefore, in the image forming apparatus **101**, it is possible to suppress the power consumption in the image forming apparatus **101** depending on whether or not the USB host **407** is connected and the setting state of the setting **501**, that is, according to the usage state of the image forming apparatus **101**.

[0053] In addition, in the embodiment described above, an apparatus to which the information processing apparatus can be applied is an image forming apparatus, but is not limited to be an image forming apparatus and may be, for example, a desktop type or notebook type personal computer, a tablet terminal, a smartphone, or the like.

OTHER EMBODIMENTS

[0054] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0055] While the present invention has been described with reference to example embodiments, it is to be understood that the invention is not limited to the disclosed example embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0056] This application claims the benefit of Japanese Patent Application No. 2024-031146, filed on Mar. 1, 2024.

Claims

1. An information processing apparatus which is configurable in a first power state in which the information processing apparatus is supplied with power, and a second power state in which power consumption is lower than that of the first power state, wherein the information processing apparatus, to which a USB device is attachably and detachably connected, comprises: a power control unit configured to switch between the first power state and the second power state; an operation section configured to perform a selection operation that selects enabling the use of the USB device by the information processing apparatus or disabling the use of the USB device by the information processing apparatus; and a determining unit configured to determine whether the information processing apparatus is in a connected state in which the USB device is connected to the information processing apparatus, or in an unconnected state in which the USB device is not connected to the information processing apparatus, and wherein the determining unit is configured, in a case where the USB device is connected to the information processing apparatus and the use of the USB device by the information processing apparatus is possible and disabling the use of the USB device has been selected by the operation section, to determine that the information

processing apparatus is in the unconnected state, and the power control unit is configured, in a case where the determining unit determines that the information processing apparatus is in the unconnected state, to switch to the second power state.

2. The information processing apparatus according to claim 1, wherein the determining unit is configured, in a case where the USB device is not currently connected to the information processing apparatus, to determine that the information processing apparatus is in the unconnected state.

3. The information processing apparatus according to claim 1, wherein the power control unit is configured, after a predetermined period of time has elapsed since it is determined by the determining unit that the information processing apparatus is in the unconnected state, to switch to the second power state.

4. The information processing apparatus according to claim 1, wherein the determining unit is configured, in a case where the USB device is connected to the information processing apparatus so that the use of the USB device by the information processing apparatus is possible and enabling use of the USB device has been selected by the operation section, to determine that the information processing apparatus is in the connected state.

5. The information processing apparatus according to claim 4, wherein the power control unit is configured, in a case where the determining unit determines that the information processing apparatus is in the connected state, to switch to the first power state.

6. The information processing apparatus according to claim 1, wherein when determining whether the information processing apparatus is in the connected state or in the unconnected state, prior to the determination, the determining unit is configured to wait for a predetermined period of time.

7. The information processing apparatus according to claim 6, wherein the determining unit is configured, after the predetermined period of time has elapsed, to determine whether the information processing apparatus is in the connected state or in the unconnected state.

8. The information processing apparatus according to claim 1, wherein the determining unit is capable of determining the presence or absence of power being supplied to the USB device.

9. The information processing apparatus according to claim 8, wherein the determining unit is configured, in a case where disabling the use of the USB device has been selected by the operation section, to omit the determination of the presence or absence of power being supplied to the USB device, and determine that the information processing apparatus is in the unconnected state.

10. The information processing apparatus according to claim 8, wherein the determining unit is configured, in a case where enabling the use of the USB device has been selected by the operation section and it is determined that no power is supplied to the USB device, to determine that the information processing apparatus is in the unconnected state.

11. The information processing apparatus according to claim 1, further comprising: an information generating unit configured to, in a case where enabling use of the USB device has been selected by the operation section, generate information indicating that use of the USB device by the information processing apparatus is possible, and wherein the determining unit is configured, based on the information generated by the information generating unit, to determine that use of the USB device by the information processing apparatus is possible.

12. The information processing apparatus according to claim 1, further comprising: a display unit configured to display a screen, and wherein the operation section is displayed on the display unit as an operation screen on which the selection operation is performed.

13. The information processing apparatus according to claim 1, wherein the second power state is a shutdown state.

14. The information processing apparatus according to claim 1, wherein the information processing apparatus is an image forming apparatus.

15. A control method for controlling an information processing apparatus that is configurable in a first power state in which the information processing apparatus is supplied with power and a

second power state in which power consumption is lower than that of the first power state, wherein the information processing apparatus is configured to be attachably and detachably connected to a USB device, the control method comprising: a power control step of performing switching control for switching between the first power state and the second power state; an operation step for performing a selection operation that selects enabling the use of the USB device by the information processing apparatus or disabling the use of the USB device by the information processing apparatus; and a determining step for determining whether the information processing apparatus is in a connected state in which the USB device is connected to the information processing apparatus or in an unconnected state in which the USB device is not connected to the information processing apparatus, and wherein in the determining step, in a case where the USB device is connected to the information processing apparatus and use of the USB device by the information processing apparatus is possible and disabling the use of the USB device has been selected in the operation step, it is determined that the information processing apparatus is in the unconnected state, and in the power control step, in a case where the determining step determines that the information processing apparatus is in the unconnected state, it is switched to the second power state.

16. A non-transitory computer-readable storage medium storing a program for causing a computer to execute a control method for controlling an information processing apparatus that is configurable in a first power state in which the information processing apparatus is supplied with power and a second power state in which power consumption is lower than that of the first power state, wherein the information processing apparatus is configured to be attachably and detachably connected to a USB device, the control method comprising: a power control step of performing switching control for switching between the first power state and the second power state; an operation step for performing a selection operation that selects enabling the use of the USB device by the information processing apparatus or disabling the use of the USB device by the information processing apparatus; and a determining step of determining whether the information processing apparatus is in a connected state in which the USB device is connected to the information processing apparatus or in an unconnected state in which the USB device is not connected to the information processing apparatus, and wherein in the determining step, in a case where the USB device is connected to the information processing apparatus and use of the USB device by the information processing apparatus is possible and disabling the use of the USB device has been selected in the operation step, it is determined that the information processing apparatus is in the unconnected state, and in the power control step, in a case where the determining step determines that the information processing apparatus is in the unconnected state, it is switched to the second power state.
